

EC 111: Principles of Microeconomics

Lecture Notes, Part I (Chapters 1-4)

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1 Ten Principles of Economics

The word *economy* comes from the Greek *oikonomos*, “one who manages a household.” The comparison is exact. A household decides who cooks, who takes the job, whether to eat out or stay in, and it makes each choice with a limited budget and a limited number of hours. A society faces the same problem at scale: who grows the food, who writes the software, who gets the apartment near the train station. In both cases the wants are many and the means are limited.

Definition 1.1 (Scarcity). The limited nature of society’s resources. Because resources are scarce, a society cannot produce everything its members would like to have.

Definition 1.2 (Economics). The study of how society manages its scarce resources.

In most economies nobody manages those resources from the top. They are allocated by the combined choices of millions of households and firms. Economics is not the study of money. It is the study of choice under constraints.

Two questions from the first day set the tone for the course.

Example 1.1 (Uber on New Year’s Eve). Shortly after midnight on January 1, ride requests spike in every large city and Uber’s price multiplier climbs to several times the normal fare. Some riders wait an hour, share a car, or walk. Drivers who were about to log off stay out, and some at home go out. The high price does what no dispatcher could do in real time: it moves scarce cars to the people who value them most and pulls more cars onto the road. Whether this is *fair* is a separate and legitimate question. The narrower point is that a price is information and an incentive at once.

Example 1.2 (Why did you choose to go to college?). The usual answers (a better job, higher pay, what the family expected) are all benefits. Economics insists on the other half of the ledger: what did you give up? Most students underestimate it because the largest item never shows up on a bill. Principle 2 puts a number on it.

The ten principles fall in three groups: how people make decisions, how people interact, and how the economy as a whole works.

1.1 How people make decisions

Principle 1: People face trade-offs

To get one thing you want, you usually give up another thing you want. A student who goes out the night before an exam has less time to study. A worker who takes more shifts has more income and less leisure. A firm that opens a new plant cannot use the same money to expand the old one.

Societies face trade-offs too. Three are classic: **guns versus butter** (a dollar spent on defense is not spent on consumer goods); **a clean environment versus income** (pollution rules improve health but raise the cost of producing things, so firms earn less, pay less, or charge more); and **efficiency versus equality**.

Definition 1.3 (Efficiency). Society gets the most it can from its scarce resources.

Definition 1.4 (Equality). Economic prosperity is distributed evenly among society's members.

Think of the economy as a pie. Efficiency is the size of the pie; equality is how evenly it is sliced. Taxes and transfers make the slices more equal, but they also reduce the reward for working and investing, so the pie can shrink. How large that effect is, and whether it is worth it, is an empirical and a political question. Recognizing a trade-off does not tell you which side to choose. It only guarantees that you see both sides.

Principle 2: The cost of something is what you give up to get it

Every decision requires comparing the benefits of an action with its costs. The catch is that the relevant cost is often not the one on the receipt. There is no such thing as a free lunch: a lunch paid for by someone else still costs the hour you could have spent otherwise.

Definition 1.5 (Opportunity cost). Whatever must be given up to obtain an item: money paid out plus the value of the best alternative use of the time and resources involved.

Example 1.3 (The opportunity cost of a year at Bentley). Suppose tuition and fees are \$60,000 and books are \$1,000. A student who instead worked full time without a degree could earn about \$35,000. Room and board of \$19,000 looks like a cost of college but mostly is not: you would have to eat and sleep somewhere anyway. Only the amount by which campus housing exceeds what you would otherwise have spent counts.

$$\text{Opportunity cost of one year} = 60,000 + 1,000 + 35,000 = \$96,000$$

plus any excess housing cost. The largest item, forgone earnings, never appears on an invoice.

Example 1.4 (A Saturday night). You planned to spend Saturday night in the library, which you value at \$25. A friend invites you skiing. Equipment rental is \$35 and a lift ticket is \$60. You also own a ski jacket you bought last year for \$180. The opportunity cost of skiing is

$$35 + 60 + 25 = \$120.$$

The jacket is irrelevant: its \$180 is gone whether you ski or study.

Common mistake

Two errors show up on every exam: counting room and board in full as a cost of college, and including a cost already paid, such as last year's jacket, in a decision made today. The test is always the same: does this amount change depending on what I choose now? If not, leave it out.

Principle 3: Rational people think at the margin

Rational people systematically and purposefully do the best they can to achieve their objectives, given the opportunities they have. They rarely face all-or-nothing choices. The question is not “should I study or not” but “should I study one more hour.” Decisions like this are about marginal changes.

Definition 1.6 (Marginal change). A small incremental adjustment to a plan of action. The *marginal benefit* is the additional benefit from that adjustment; the *marginal cost* is the additional cost.

The rule: take the action if and only if its marginal benefit is at least as large as its marginal cost. Keep going while $MB \geq MC$; stop when $MB < MC$. Marginal benefit usually falls as you do more of something (the fifth slice of pizza is worth less than the first) and marginal cost usually rises (the fifth hour of studying is harder than the first). The best quantity is where the two curves cross (Figure 1).

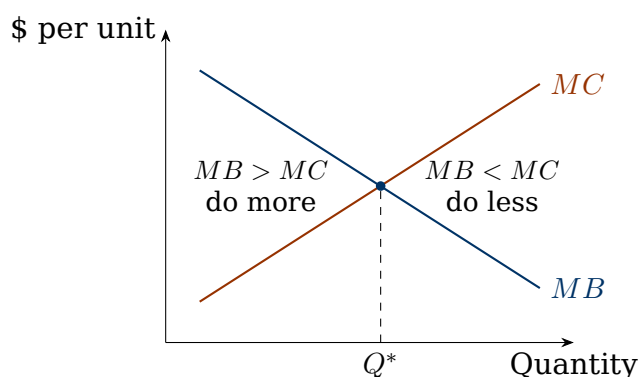


Figure 1: Thinking at the margin. Expand the activity while marginal benefit exceeds marginal cost; stop at Q^* .

Example 1.5 (The unfinished product). Your company has spent \$5 million developing a new product. Competitors have just launched similar products, so expected sales of yours are now \$3 million. Finishing development and producing the product would cost \$1 million. Should you go ahead?

Yes. The marginal benefit of finishing is \$3 million and the marginal cost is \$1 million. The \$5 million is sunk and does not enter the decision. The project as a whole loses \$3 million (\$6 million spent against \$3 million in sales), but stopping now loses \$5 million. The most you should pay to finish is \$3 million.

Definition 1.7 (Sunk cost). A cost that has already been committed and cannot be recovered. Rational decisions ignore sunk costs.

Marginal thinking resolves an old puzzle. Water is essential and cheap; diamonds are useless for survival and expensive. The marginal cup of water is worth little because water is plentiful; the marginal diamond is worth a lot because diamonds are rare. Prices reflect marginal, not total, value.

Principle 4: People respond to incentives

Definition 1.8 (Incentive). Something that induces a person to act, such as the prospect of a reward or a punishment.

Since rational people compare marginal benefits and marginal costs, anything that changes those benefits or costs changes behavior. Raise the price of doughnuts and consumers buy fewer while bakers make more. Predicting these responses is the core of what economists do.

Example 1.6 (A gasoline tax). The government raises the gasoline tax by \$1 per gallon. Consumers drive less, carpool, switch to smaller or electric cars, move closer to work, and use transit more. Automakers shift production toward fuel-efficient models. Trucking firms raise their rates. Nobody was ordered to do any of this. The tax changed the price, and the price changed the calculation. A surge multiplier on New Year's Eve works the same way: a raise for drivers and a price increase for riders, and both respond.

Incentives also produce *unintended consequences*. Seat belt laws make accidents less costly for the driver, so careful driving pays less. Peltzman (1975) found fewer deaths per accident but more accidents, and more pedestrian deaths. A policy that ignores incentives will be surprised by the outcome.

Key point

Principles 1 through 4 are one idea seen from four sides. Scarcity forces trade-offs; the cost of a choice is the best alternative forgone; the right comparison is at the margin; and when marginal costs or benefits change, behavior changes.

1.2 How people interact

Principle 5: Trade can make everyone better off

If each person had to grow their own food, sew their own clothes, and cut their own hair, we would all be poorer, hungrier, and worse dressed. Specialization lets each person do what they do relatively well and trade for the rest. Trade is not a contest with a winner and a loser: two parties agree to a voluntary exchange only if each expects to gain.

The same logic applies between countries. Trade between the United States and China lets each specialize in what it produces at lower cost and gives consumers in both more variety at lower prices. Not everyone within each country gains; some workers and firms lose. Chapter 3 works this out in detail.

Principle 6: Markets are usually a good way to organize economic activity

Definition 1.9 (Market). A group of buyers and sellers of a particular good or service. They need not be in one place.

Every economy must answer three questions: what to produce, how to produce it, and who gets it. In a centrally planned economy a government office answers them. In a **market economy**, which allocates resources through the decentralized decisions of many firms and households, nobody does. Households decide where to work and what to buy; firms decide whom to hire and what to make; each acts in its own interest. The remarkable fact, first stated by Adam Smith in 1776, is that this decentralized process usually produces good outcomes, as if guided by an *invisible hand*.

Prices are the mechanism. A price reflects both what buyers are willing to pay and what it costs to produce. When buyers want more of something, its price rises, which tells producers to make more and buyers to economize. No one has to be told anything.

Example 1.7 (Taxis and Uber). For decades cities fixed the number of taxis through medallions and fixed the fares by law. The results were long waits at peak times and medallions that sold for over a million dollars in New York. Uber, launched in 2009, let the price move. Surge pricing is the invisible hand made visible: when demand spikes, the price rises, more drivers come out, and rides go to those who value them most. Taxi owners were not pleased. Consumers were.

Principle 7: Governments can sometimes improve market outcomes

Markets need governments in one obvious way: someone has to enforce the rules.

Definition 1.10 (Property rights). The ability of an individual to own and exercise control over scarce resources.

A farmer will not plant if the harvest can be stolen; a firm will not invent if its invention can be copied freely. Police and courts make markets possible. Beyond enforcing rules, governments intervene to promote efficiency and equality.

Definition 1.11 (Market failure). A situation in which a market left on its own fails to allocate resources efficiently.

Definition 1.12 (Externality). The impact of one person's actions on the well-being of a bystander. A factory and its customers do not pay for the health costs its pollution imposes on the neighbors.

Definition 1.13 (Market power). The ability of a single economic actor, or a small group, to have a substantial influence on market prices. The only gas station within 60 miles restricts output to keep its price high.

The invisible hand also does not guarantee that everyone has enough. A market rewards people according to their ability to produce things others will pay for, and abilities differ. The income tax and the welfare system exist to change how the pie is divided.

Example 1.8 (OxyContin, 1996–2010). In 1996 Purdue Pharma released OxyContin, a slow-release opioid painkiller, and marketed it as low-risk. The market failed on two counts: patients and doctors were misinformed about addiction, and the cost of abuse fell on families, hospitals, and communities rather than on the seller. In 2007 the company pleaded guilty to misbranding and paid over \$600 million. In August 2010 the FDA approved an abuse-deterrent reformulation that made the pills hard to crush or dissolve.

The regulation then failed too. Users who could no longer abuse OxyContin switched to heroin, which was cheaper and widely available. Heroin deaths, flat for a decade, more than tripled between 2010 and 2014, most in the states where OxyContin abuse had been highest. Evans, Lieber, and Power (2019) estimate that the reformulation explains a large share of the rise and that total opioid deaths did not fall. Markets fail, and regulation that ignores Principle 4 fails as well.

Key point

Principle 7 says governments *can* improve market outcomes, not that they always do. A policy that changes prices changes behavior in ways its designers did not intend. Whether an intervention helps is answered case by case, with evidence.

1.3 How the economy as a whole works

The last three principles belong to macroeconomics, which is not the subject of this course, but they complete the picture.

Principle 8: A country's standard of living depends on its ability to produce goods and services. Average income in the United States is roughly ten times that of Nigeria and eight times what it was a century ago. Almost all of the difference is **productivity**, the quantity of goods and services produced from each unit of labor input: more equipment, better training, better technology. To raise living standards, raise productivity.

Principle 9: Prices rise when the government prints too much money. Inflation is an increase in the overall level of prices. In Germany a newspaper cost 0.3 marks in 1921 and 70 million marks by late 1922; the quantity of money had grown at the same pace. High inflation is almost always the result of rapid growth in the quantity of money.

Principle 10: Society faces a short-run trade-off between inflation and unemployment. In the long run, printing money raises only prices. In the short run, more money means more spending, more production, more hiring, and less unemployment. Policies that reduce inflation therefore tend to raise unemployment for a while, and the reverse.

In class

Come to class with an answer to “why did you choose to go to college,” stated as a cost and a benefit, and explain it to one other person. Kristensen and Bjerkedal (*Science*, 2007) found that firstborns score higher on cognitive tests than their siblings, and that the advantage belongs to whoever *acts* as the eldest, not to birth order. The eldest spends years explaining things to the younger ones. Teaching consolidates learning; study groups do the same for you.

Exercises

Exercise 1.1. Maya is deciding whether to attend a one-year graduate program. Tuition is \$45,000 and books and fees are \$2,000. Her current job pays \$52,000 a year. She would pay \$14,000 for an apartment near campus; she currently pays \$12,000 in rent. Last year she spent \$800 on a test-prep course to get admitted.

- (a) Compute the opportunity cost of the year in the program.
- (b) Which of the listed amounts are irrelevant to her decision, and why?

Exercise 1.2. You run a food truck. Three months ago you paid a nonrefundable \$3,000 permit fee to sell at a two-day festival. The forecast now calls for rain. If you attend, you expect \$1,800 in revenue and will spend \$1,200 on ingredients and a helper’s wages. Staying home earns nothing and costs nothing.

- (a) Should you attend? Show the marginal comparison.
- (b) Suppose expected revenue falls to \$1,000. Should you attend?
- (c) A friend says, “You already paid \$3,000, you have to go.” What is wrong with this argument?

Exercise 1.3. Boston introduces a \$5 charge for driving into downtown between 7 and 10 a.m. on weekdays.

- (a) List three ways commuters might respond and two ways businesses might respond.
- (b) Would the true cost of driving downtown at 8 a.m. rise by exactly \$5? Explain using opportunity cost.

Answers to exercises**Exercise 1.**

- (a) Tuition, books, and fees \$47,000, plus forgone salary \$52,000, plus the extra housing cost of \$2,000 (\$14,000 minus the \$12,000 she pays anyway). Opportunity cost = $47,000 + 52,000 + 2,000 = \$101,000$.
- (b) The \$800 test-prep course is sunk: it was paid last year and is not recoverable whatever she decides. The \$12,000 of current rent is not a cost of the program either; only the \$2,000 difference is.

Exercise 2.

- (a) Marginal benefit \$1,800, marginal cost \$1,200. Attend, since $1,800 > 1,200$; you net \$600 relative to staying home.
- (b) Marginal benefit \$1,000 is less than marginal cost \$1,200. Stay home; attending would lose a further \$200.
- (c) The \$3,000 is sunk. It is gone whether you attend or not, so it cannot be a reason to do either. The only question is whether attending adds more revenue than cost.

Exercise 3.

- (a) Commuters: take the T, shift the trip to before 7 or after 10, carpool, work from home, move closer to the office. Businesses: move deliveries outside the window, offer flexible start times, relocate outside the zone, raise prices to cover delivery costs.
- (b) No. If the charge reduces traffic, the trip is faster, so its time cost falls. The full opportunity cost (money plus time) rises by less than \$5 and could fall for someone whose time is very valuable.

2 Thinking Like an Economist

Chapter 1 listed the principles. This chapter is about method: how economists turn a messy world into something they can analyze. You will meet the first two models of the course, the circular-flow diagram and the production possibilities frontier (PPF), and you will learn the difference between a statement that can be tested and one that cannot. The appendix reviews the graphing tools used for the rest of the semester.

2.1 The economist as scientist

Economists wear two hats. As *scientists*, they try to explain how the economy works. As *policy advisers*, they try to improve it. The two jobs use different kinds of statements, and keeping them apart is the main lesson of this chapter.

2.1.1 The scientific method

The economist's method is the same one used in physics or biology: observe, build a theory that could explain what you saw, then look at new data to see whether the theory survives. The difference is practical. A chemist can run a controlled experiment; an economist who wants to know what a higher minimum wage does to employment cannot ask Congress to raise the wage in half the country and hold it fixed in the other half. Economists instead rely on the *natural experiments* that history provides: one state raises its minimum wage while the neighboring state does not, and the two can be compared.

2.1.2 Assumptions

Every science simplifies. A physicist studying a falling marble ignores air resistance because it barely matters; the same assumption would be foolish for a falling feather. Economists do the same thing. To study international trade we may assume the world has two countries and two goods. There are about 200 countries and millions of goods, but the two-by-two world captures the logic of trade, and the logic does not change when the numbers grow.

Key point

The art is choosing the assumption that fits the question. For a question about prices next month, assuming prices are fixed is reasonable. For a question about prices over a decade, it is not.

2.1.3 Models

Definition 2.1 (Economic model). A simplified representation of some part of the economy, built from assumptions, and expressed in words, diagrams, or equations.

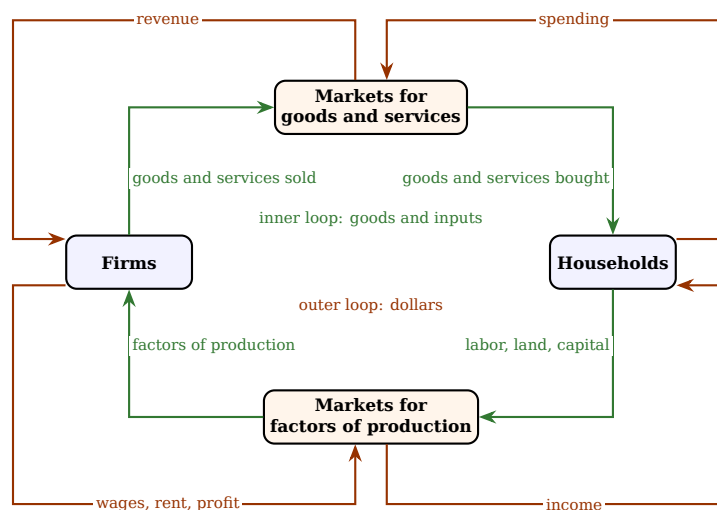
A plastic model of the human body leaves out most of the body's detail and is more useful for it: you can see where the organs sit. Economic models leave out detail for the same reason. A model is judged by whether it answers the question at hand, not by how much of reality it reproduces.

2.2 The circular-flow diagram

Definition 2.2 (Circular-flow diagram). A visual model of the economy that shows how dollars flow through markets between households and firms.

The model has two kinds of decision makers and two kinds of markets.

- **Households** own the factors of production (labor, land, capital) and sell or rent them to firms. They use the income to buy goods and services.
- **Firms** hire the factors of production and use them to produce goods and services, which they sell to households.
- In the **market for goods and services**, firms are sellers and households are buyers.
- In the **market for factors of production**, households are sellers and firms are buyers.



The diagram has two loops. The inner loop shows real things: inputs move from households to firms, and goods and services move from firms to households. The outer loop shows dollars moving in the opposite direction: households spend on goods, that spending is revenue to firms, firms pay for inputs, and those payments are income to households.

Example 2.1 (Following a dollar). You pay \$5 for a coffee: spending in the market for goods and services, revenue to the cafe. The cafe pays it out as the barista's wage, the landlord's rent, and the owner's profit. All three are households, so the \$5 is now income, and tomorrow it is spent again.

Note

The model leaves out the government, the financial system, and the rest of the world. That is deliberate. Adding them makes the diagram more realistic and, for the questions we ask now, no more useful.

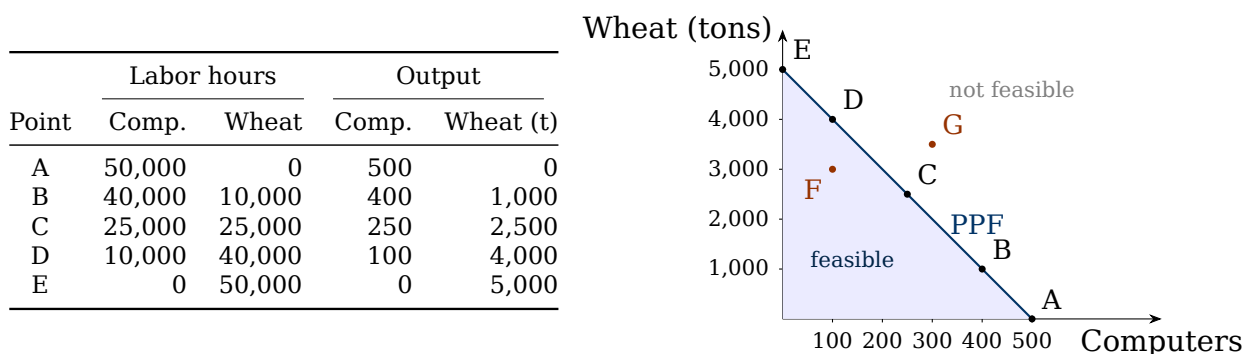
2.3 The production possibilities frontier

Definition 2.3 (Production possibilities frontier). A graph showing the combinations of output that an economy can produce given its available factors of production and its technology.

We build the PPF from a small economy with one resource and two goods.

- The only resource is labor: 50,000 hours per month.
- One computer takes 100 hours of labor. One ton of wheat takes 10 hours.
- Using all 50,000 hours on computers yields $50,000/100 = 500$ computers. Using all of them on wheat yields $50,000/10 = 5,000$ tons.

Every split of the 50,000 hours between the two industries gives one output combination. The table lists five; the graph plots them. Check one row: at C, 25,000 hours on computers produce $25,000/100 = 250$ computers, and the other 25,000 hours produce $25,000/10 = 2,500$ tons of wheat.



The line through A to E is the PPF. Because the two goods use labor at fixed rates, the frontier is a straight line. The graph divides all output combinations into three groups.

- **On the frontier** (A through E): efficient. Every hour is used, and the only way to get more of one good is to give up some of the other.
- **Inside the frontier**: feasible but inefficient. Some labor is idle or badly used, and the economy could have more of both goods.
- **Outside the frontier**: not feasible with current resources and technology.

Example 2.2 (Points F and G). Point F is 100 computers and 3,000 tons of wheat. It needs $100 \times 100 + 3,000 \times 10 = 10,000 + 30,000 = 40,000$ hours. That is less than 50,000, so F is feasible, but 10,000 hours are unused. F lies inside the frontier. Moving from F straight up to D (100 computers, 4,000 tons) or straight right to (200 computers, 3,000 tons) uses the idle hours and gives more of one good at no cost in the other.

Point G is 300 computers and 3,500 tons. It needs $300 \times 100 + 3,500 \times 10 = 30,000 + 35,000 = 65,000$ hours. The economy has 50,000. G is outside the frontier and cannot be produced.

In class

Pick a combination of computers and wheat. Compute the labor it requires and say where the point lies. The rule is one line: hours needed = $100 \times \text{computers} + 10 \times \text{wheat}$, compared with 50,000.

2.3.1 Opportunity cost and the slope of the PPF

Moving along the frontier means moving labor from one industry to the other. Going from A to B, 10,000 hours leave computers and go to wheat: the economy loses 100 computers and gains 1,000 tons. Going from B to C, 15,000 hours move: 150 fewer computers, 1,500 more tons. The exchange rate never changes. Each computer given up frees 100 hours, which grow 10 tons of wheat.

Key point

Along this PPF the opportunity cost of one computer is 10 tons of wheat, and the opportunity cost of one ton of wheat is $1/10$ of a computer. The opportunity cost of the good on the horizontal axis is the absolute value of the slope of the PPF: $5,000/500 = 10$.

Common mistake

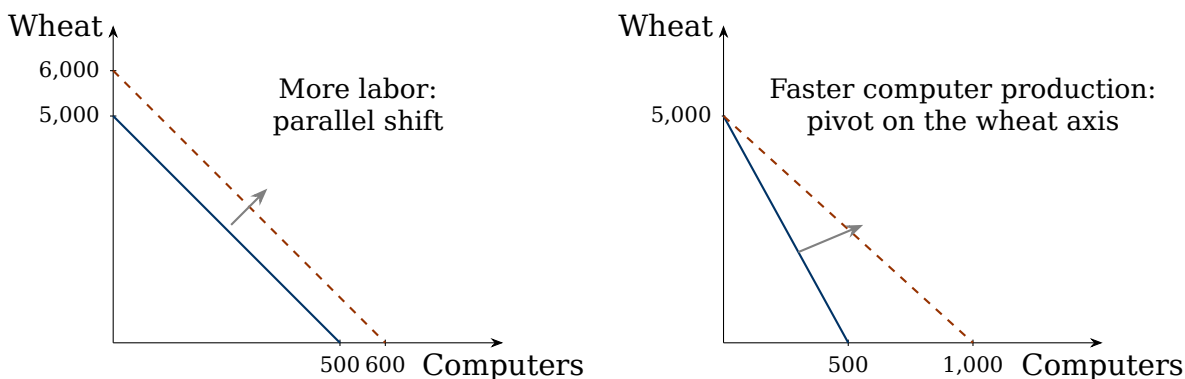
The slope tells you the opportunity cost of the horizontal-axis good in units of the vertical-axis good. If the axes were swapped, the slope would be $1/10$. Always say which good you are pricing and what it is priced in.

2.3.2 Shifts of the PPF

The frontier is drawn for a fixed amount of labor and a fixed technology. Change either and the frontier moves.

More resources. Suppose immigration adds 10,000 labor hours per month, so the economy has 60,000. The all-computer point becomes $60,000/100 = 600$ and the all-wheat point $60,000/10 = 6,000$. Both intercepts rise by 20 percent, and the new frontier is parallel to the old one. Opportunity cost is unchanged: a computer still costs 10 tons. Point G (300, 3,500) needed 65,000 hours and is still out of reach.

Better technology. Suppose instead that a new process cuts the labor per computer from 100 hours to 50, with wheat unchanged. All 50,000 hours in computers now yield 1,000; all in wheat still yield 5,000. The frontier pivots on the wheat intercept and swings out along the computer axis. Because the computer axis moved and the wheat axis did not, the slope changes: a computer now frees 50 hours, which grow 5 tons, so the opportunity cost of a computer falls from 10 tons to 5.



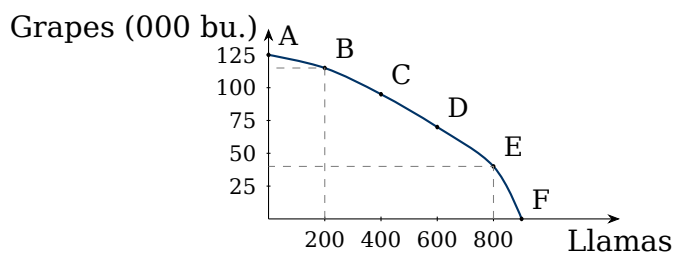
Note

Each of these is economic growth: the set of feasible combinations expands. Growth shifts the frontier; it is not a move along it. Chapter 3 uses the same linear PPF to show how two countries with different slopes gain from trade.

2.3.3 A bowed-out PPF

The computer-and-wheat frontier is straight because every labor hour is identical. Most resources are not. Suppose you own land in California and can use it to grow grapes or raise llamas. Some of the land is fertile and grows excellent grapes; the rest is rocky and poor for vines. Llamas do equally well anywhere. The frontier is bowed outward because opportunity cost rises as more of a good is produced.

Point	Llamas	Grapes (000 bu.)	Cost per llama (bushels)
A	0	125	
B	200	115	50
C	400	95	100
D	600	70	125
E	800	40	150
F	900	0	400



Start at A with only grapes. The first 200 llamas go on the rocky land, which was contributing little to the harvest, so grape output falls by only 10 thousand bushels: 50 bushels per llama. Each further 200 llamas takes better land, and the cost rises: 20, 25, 30 thousand bushels. The last 100 llamas displace the best vineyard and cost 40 thousand bushels, or 400 bushels per llama. The frontier is bowed outward because opportunity cost rises as more of a good is produced.

Key point

A straight PPF means constant opportunity cost; a bowed-out PPF means increasing opportunity cost. The bow comes from resources that are not equally suited to both uses: the first units of a good use the resources with the lowest opportunity cost, and later units use resources that were more valuable in their old job.

2.4 Microeconomics and macroeconomics

Definition 2.4 (Microeconomics). The study of how households and firms make decisions and how they interact in markets.

Definition 2.5 (Macroeconomics). The study of economy-wide phenomena, including inflation, unemployment, and economic growth.

A microeconomist asks how a tariff on steel changes the price of cars, or why rents in Boston rose. A macroeconomist asks why the whole economy grows in some decades and stalls in others. The two are connected, since the whole economy is the sum of many markets, but they use different tools and ask different questions. This course is the micro half.

2.5 The economist as policy adviser

When an economist explains why rents rose, she is a scientist. When she says what the city should do about it, she is a policy adviser. The two activities produce different kinds of claims.

Definition 2.6 (Positive statement). A claim about how the world is. It can, in principle, be confirmed or refuted by evidence.

Definition 2.7 (Normative statement). A claim about how the world ought to be. It rests on values as well as facts and cannot be settled by evidence alone.

“A \$20 minimum wage would reduce teen employment” is positive. It might be wrong, but data could show that. “The minimum wage should be \$20” is normative. Evidence on employment matters to it, but so does a judgment about whose welfare counts and how much. No dataset can settle the second question.

Common mistake

Positive does not mean true. “The moon is made of cheese” is a positive statement: testable and false. The test for positive versus normative is whether evidence *could* settle it, not whether the claim is correct. Words like “should,” “ought,” and “needs to” usually mark a normative claim.

In class

Positive or normative?

- (a) Prices rise when the central bank prints more money.
- (b) The central bank should print less money.
- (c) A tax cut is needed to get the economy moving.
- (d) A higher price of burritos raises the demand for ramen.

(a) and (d) are positive: each describes a relationship that data could test. (b) and (c) are normative: “should” and “needed” express a judgment about goals.

Economists advise throughout government (the Council of Economic Advisers, the Treasury, the Congressional Budget Office, the Federal Reserve), but their advice is one input among many. The political process, not the economist, makes the final call.

2.6 Why economists disagree

Two economists on a news panel take opposite sides on trade, and the viewer concludes that economists know nothing. There are two better explanations.

1. **Different scientific judgments.** Economists can disagree about which positive theory is right, or about the size of an effect: how much does employment fall when the minimum wage rises by a dollar? These are empirical questions, and the evidence is sometimes not yet decisive.
2. **Different values.** Two economists can agree on every fact and still disagree about what to do, because they weigh the winners and losers differently. That is a normative disagreement, and no amount of data resolves it.

A third explanation is that the disagreement is smaller than it looks. In surveys, about 93 percent of economists agree that rent ceilings reduce the quantity and quality of housing, and 93 percent that tariffs and import quotas usually reduce general welfare. A producer looking for balance will still find one economist on each side.

2.7 Appendix: graphing

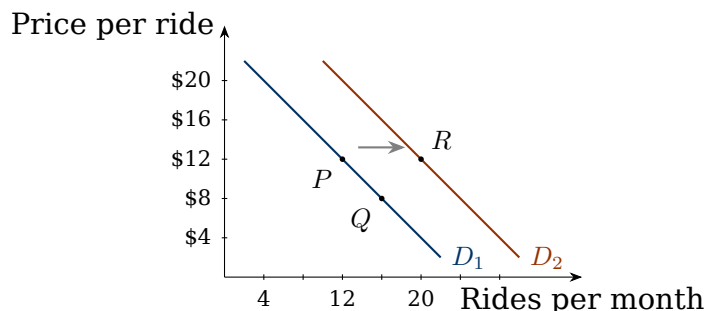
Most of the models in this course are graphs. This appendix collects the rules.

2.7.1 Curves in the coordinate system

A graph with two variables puts one on the horizontal (x) axis and one on the vertical (y) axis; each pair of values is a point. A curve is the set of points a relationship generates while everything else is held fixed. When y rises with x the curve slopes up (positive relationship); when y falls as x rises it slopes down (negative relationship).

Example 2.3 (A demand curve). Maya's rides per month with a ride-hailing app depend on the price per ride and on her monthly income. By convention price goes on the vertical axis and quantity on the horizontal axis. Each column of the table is a curve: D_1 is her demand curve at \$2,000 of income, D_2 at \$3,000.

Price per ride	Rides per month	
	income \$2,000	income \$3,000
\$4	20	28
\$8	16	24
\$12	12	20
\$16	8	16
\$20	4	12



2.7.2 Moving along a curve versus shifting it

The distinction is the most important graphing rule in the course.

- A change in a variable that is *on an axis* moves you *along* the curve. If the price falls from \$12 to \$8 with income fixed at \$2,000, Maya moves from P (12 rides) to Q (16 rides) along D_1 .
- A change in a variable that is *not on an axis* *shifts* the whole curve. If her income rises to \$3,000, at every price she takes 8 more rides: the curve moves right, from D_1 to D_2 , and at \$12 she goes from P to R .

Common mistake

"Price fell, so demand increased" is wrong in this vocabulary. A lower price raises the *quantity demanded* (a move along the curve). "Demand increased" means the curve shifted, which requires something other than the good's own price to change.

2.7.3 Slope

Definition 2.8 (Slope). The ratio of the vertical change to the horizontal change between two points on a curve: rise over run, $\Delta y / \Delta x$.

Between P and Q on D_1 , price falls by \$4 and rides rise by 4, so the slope is $-4/4 = -1$. A negative slope means the curve falls to the right. A steep curve means y responds a lot to a small change in x ; a flat curve means it barely responds. On the computer-and-wheat PPF the slope was $-5,000/500 = -10$: each extra computer costs 10 tons of wheat. Slope is the graph's way of showing a trade-off.

2.7.4 Cause and effect

Graphs show that two variables move together. They do not by themselves show that one causes the other. Two traps recur.

Omitted variables. Ice cream sales and drownings rise and fall together across the months of the year. Ice cream does not cause drowning; hot weather raises both. A graph of the two would show a clear positive relationship and prove nothing about cause. Before reading a cause off a graph, ask what third variable might drive both.

Reverse causality. Cities with more police officers tend to have more crime. Do police cause crime? More likely, cities with a crime problem hire more police: the causation runs from y to x , not from x to y . Timing does not settle it either, since the police may be hired in anticipation of crime. Only a controlled experiment, or a natural experiment that moves one variable for reasons unrelated to the other, separates cause from correlation.

Key point

Correlation is what a graph can show. Causation is a claim about what would happen if you changed one variable and held everything else fixed. The gap between the two is where most bad economic arguments live.

Exercises

Exercise 2.1. Use the computer-and-wheat economy: 50,000 labor hours, 100 hours per computer, 10 hours per ton of wheat.

- (a) Point H is 200 computers and 2,800 tons of wheat. How many hours does it require? Is it on, inside, or outside the frontier?
- (b) Starting from H, what is the largest wheat output the economy could reach without producing fewer computers?
- (c) The economy moves from B (400, 1,000) to C (250, 2,500). What is the opportunity cost per computer given up? Does it match the slope?
- (d) A new seed cuts the labor per ton of wheat from 10 hours to 5, with computers unchanged. Give the new intercepts, describe the shift, and compute the new opportunity cost of a computer.

Exercise 2.2. Use the grape-and-llama land from the table above.

- (a) What is the opportunity cost of raising llamas from 400 to 600? From 600 to 800? State each in bushels per llama.
- (b) Why does the cost rise? What feature of the land drives the answer?
- (c) Suppose every acre were equally good for grapes and for llamas. What shape would the frontier take, and why?

Exercise 2.3. (a) Positive or normative? (i) Bentley should lower tuition. (ii) U.S. inflation was 3 percent last year. (iii) Free trade lowers wages for factory work-

- ers. (iv) The Federal Reserve ought to cut interest rates before the election. (v) Rent control in Cambridge reduced the supply of apartments.
- (b) Students who sit in the front row earn higher grades on average. A dean proposes assigning seats so every student sits in front at least once. Give one omitted-variable and one reverse-causality reason the policy might not raise grades.
- (c) Maya's income falls to \$2,000 while the price stays at \$12. On the demand graph, is this a move along a curve or a shift? Where does she end up?

Answers to exercises

Exercise 2.1. (a) $200 \times 100 + 2,800 \times 10 = 20,000 + 28,000 = 48,000$ hours, less than 50,000. H is inside the frontier, with 2,000 hours idle. (b) Keep 20,000 hours on computers; put the remaining 30,000 on wheat: 3,000 tons. That is the frontier point directly above H. (c) Giving up 150 computers frees 15,000 hours, which grow 1,500 tons. $1,500/150 = 10$ tons per computer, the same as the slope of -10 (in absolute value). Along a straight PPF the cost is the same everywhere. (d) Computer intercept unchanged at 500; wheat intercept doubles to $50,000/5 = 10,000$ tons. The frontier pivots on the computer intercept and swings up along the wheat axis. A computer now costs $100/5 = 20$ tons of wheat: its opportunity cost rose, because wheat became cheaper to produce.

Exercise 2.2. (a) 400 to 600: grapes fall from 95 to 70, so 25 thousand bushels for 200 llamas, or 125 bushels per llama. 600 to 800: 70 to 40, so 30 thousand bushels, or 150 per llama. (b) Land is heterogeneous. The first llamas use land that was poor for grapes and cost little harvest. Each later batch takes land that was more productive in grapes, so each llama displaces more bushels. (c) A straight line. If every acre gave up the same bushels when converted to llamas, opportunity cost would be constant and the frontier would be linear, like the computer-and-wheat example.

Exercise 2.3. (a) (i) normative; (ii) positive; (iii) positive (testable, whether or not it is true); (iv) normative; (v) positive. (b) Omitted variable: motivated students both choose the front row and study more, so motivation, not the seat, raises grades. Reverse causality: students who are already doing well feel comfortable sitting in front, so grades drive seating rather than the other way around. Assigned seating would move the seat without moving the motivation or the prior grades. (c) Income is not on either axis, so the curve shifts: from D_2 back to D_1 . At \$12 she moves from R (20 rides) to P (12 rides).

3 Interdependence and the Gains from Trade

Nearly everything you consume was made by people you will never meet, each of whom produced something for you and got something else in return. This is interdependence. Chapter 1 listed one of the Ten Principles: *trade can make everyone better off*. This chapter shows why, and answers three questions: How can trade make *both* sides better off, even when one side is better at producing everything? What are absolute and comparative advantage, and which one determines who should produce what? What determines the price at which two parties trade?

3.1 A simple world: two countries, two goods, one input

Two countries, the United States and Japan; two goods, computers and wheat (in tons); one input, labor, measured in hours per month.

	Labor hours	Hours per computer	Hours per ton wheat
United States	50,000	100	10
Japan	30,000	125	25

We want each country's production and consumption with and without trade.

3.1.1 Production possibilities in the United States

If the U.S. puts all 50,000 hours into computers it makes $50,000/100 = 500$ computers and no wheat. If it puts all hours into wheat it makes $50,000/10 = 5,000$ tons and no computers. Splitting labor between the two goods gives the points in between.

Because every hour converts into either good at a constant rate, the PPF is a straight line. Its slope is constant: each extra 1,000 tons of wheat costs 100 computers, so 1 ton of wheat costs $1/10$ of a computer, and 1 computer costs 10 tons of wheat.

3.1.2 Production possibilities in Japan

Japan has fewer hours and needs more hours per unit of each good. All-in on computers: $30,000/125 = 240$ computers. All-in on wheat: $30,000/25 = 1,200$ tons. Along Japan's PPF, each computer costs $125/25 = 5$ tons of wheat, and each ton of wheat costs $1/5$ of a computer.

3.1.3 Production and consumption without trade

A country that does not trade consumes exactly what it produces, so its consumption point is on its PPF. Suppose each country splits its labor evenly between the two goods.

- U.S.: 25,000 hours in each good, so 250 computers and 2,500 tons of wheat (point A, left panel of Figure 2).
- Japan: 15,000 hours in each good, so 120 computers and 600 tons of wheat (point A, right panel).

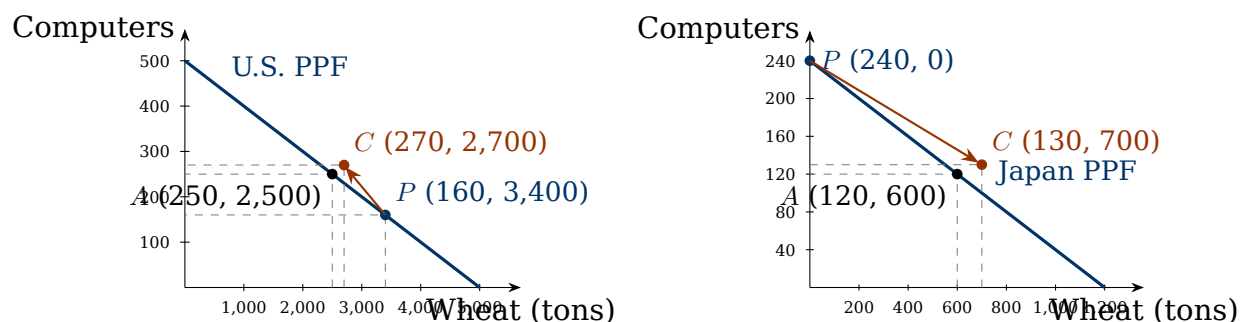


Figure 2: Left: the U.S. PPF. Right: Japan's PPF. A: production and consumption without trade. P: production with trade (Japan specializes completely in computers). C: consumption with trade, outside each PPF. The arrows are the trade: the U.S. exports 700 tons of wheat and imports 110 computers.

3.2 Specialization and trade

Now let the countries trade. Each can produce a different bundle from the one it consumes; exports and imports make up the difference.

Definition 3.1 (Exports and imports). **Exports** are goods produced at home and sold abroad. **Imports** are goods produced abroad and bought at home.

3.2.1 Production with trade

Suppose the U.S. shifts labor toward wheat and Japan shifts labor toward computers.

- The U.S. produces 3,400 tons of wheat. This takes $3,400 \times 10 = 34,000$ hours. The remaining $50,000 - 34,000 = 16,000$ hours make $16,000/100 = 160$ computers. Production point: $P = (160 \text{ computers}, 3,400 \text{ wheat})$.
- Japan produces 240 computers. This takes $240 \times 125 = 30,000$ hours, all of Japan's labor. Japan produces 0 wheat. Production point: $P = (240, 0)$.

3.2.2 Consumption with trade

Suppose the two countries agree on the following exchange: the U.S. sends 700 tons of wheat to Japan, and Japan sends 110 computers to the U.S.

	United States		Japan	
	Computers	Wheat	Computers	Wheat
Produced	160	3,400	240	0
+ Imported	110	0	0	700
– Exported	0	700	110	0
= Consumed	270	2,700	130	700

Compare with the no-trade bundles.

	Without trade	With trade	Gain
U.S. computers	250	270	+20
U.S. wheat (tons)	2,500	2,700	+200
Japan computers	120	130	+10
Japan wheat (tons)	600	700	+100

Each country ends up with more of *both* goods. The consumption points C in Figure 2 lie outside the PPFs. No country gained resources or technology; the only thing that changed is that each produced a different mix and then exchanged.

Key point

Without trade, consumption is tied to production: the consumption point must be on the PPF. With trade, production stays on the PPF but consumption can move outside it. In this sense trade works like a technological improvement: society gets bundles it could not produce on its own.

Where does the extra output come from? Add up world production. Without trade the two countries make $250 + 120 = 370$ computers and $2,500 + 600 = 3,100$ tons of wheat. With trade they make $160 + 240 = 400$ computers and $3,400 + 0 = 3,400$ tons. Reallocating labor raised world output by 30 computers and 300 tons, exactly the sum of the two countries' gains. The pie got bigger.

Note

The U.S. does not specialize completely and does not need to: Japan's whole labor force makes only 240 computers, while the two countries together consume 400. What matters is the direction of the shift. Each country moves toward the good it is relatively better at.

3.3 Absolute advantage and comparative advantage

Why did Japan specialize in computers and the U.S. in wheat, and not the other way around? Two different notions of "who is the better producer" give different answers, and only one of them is correct.

Definition 3.2 (Absolute advantage). A producer has an **absolute advantage** in a good if it can produce that good using fewer inputs than another producer.

In our example, the U.S. needs 100 hours per computer versus Japan's 125, and 10 hours per ton of wheat versus Japan's 25. The U.S. has an absolute advantage in *both* goods.

If the U.S. is better at everything, why does it import computers from Japan, and why does Japan gain? Absolute advantage measures cost in inputs. The cost that matters for the decision is opportunity cost.

The opportunity cost of one computer is the wheat that could have been produced with the labor used for that computer: hours per computer divided by hours per ton of wheat, the slope of the PPF.

- U.S.: one computer uses 100 hours, which could have made $100/10 = 10$ tons of wheat. The opportunity cost of 1 computer is 10 tons of wheat; the opportunity cost of 1 ton of wheat is $1/10$ computer.
- Japan: one computer uses 125 hours, which could have made $125/25 = 5$ tons of wheat. The opportunity cost of 1 computer is 5 tons of wheat; the opportunity cost of 1 ton of wheat is $1/5$ computer.

Opportunity cost of...	1 computer	1 ton of wheat
United States	10 tons of wheat	$1/10$ computer
Japan	5 tons of wheat	$1/5$ computer

Definition 3.3 (Comparative advantage). A producer has a **comparative advantage** in a good if it can produce that good at a lower opportunity cost than another producer.

Japan gives up 5 tons of wheat per computer; the U.S. gives up 10. Japan has the comparative advantage in computers. The U.S. gives up $1/10$ computer per ton of wheat; Japan gives up $1/5$. The U.S. has the comparative advantage in wheat.

Key point

Principle of comparative advantage. Each good should be produced by the party with the lower opportunity cost of producing it. The gains from trade come from differences in opportunity costs, not from differences in input requirements. Absolute advantage is neither necessary nor sufficient for gains from trade.

For any one producer, the opportunity cost of one good is the reciprocal of the opportunity cost of the other (10 and $1/10$; 5 and $1/5$). So a producer can have an absolute advantage in both goods but cannot have a comparative advantage in both: if your opportunity cost of computers is lower than mine, your opportunity cost of wheat must be higher. The only exception is equal opportunity costs, and then there are no gains from trade.

Common mistake

Comparing hours per unit (absolute advantage) gives the wrong answer: here it says the U.S. should make everything. Compare opportunity costs, and keep the units straight: the opportunity cost of a computer is in tons of wheat, that of wheat in computers.

3.4 The price of trade

At what rate should computers exchange for wheat? For both countries to gain, the price of a computer must lie between the two opportunity costs:

$$5 \text{ tons of wheat} < \text{price of 1 computer} < 10 \text{ tons of wheat.}$$

- Japan makes a computer at a cost of 5 tons of wheat. If it can sell a computer for more than 5 tons, it gains by exporting computers.
- The U.S. makes a computer at a cost of 10 tons of wheat. If it can buy a computer for less than 10 tons, it gains by importing computers.

Our trade was 700 tons of wheat for 110 computers, a price of $700/110 \approx 6.4$ tons of wheat per computer. That is above Japan's cost (5) and below the U.S. cost (10), so both gain. Equivalently, the price of wheat is $110/700 \approx 0.16$ computers per ton, which is between $1/10 = 0.10$ and $1/5 = 0.20$.

A price outside the range makes one side worse off. At 4 tons per computer Japan would rather grow its own wheat; at 12 the U.S. would rather build its own computers. Neither trade happens voluntarily. Where in the range the price settles depends on bargaining or, in a market, on supply and demand (Chapter 4). The closer to 10, the more of the gains go to Japan; the closer to 5, the more go to the U.S.

3.5 A second example: Argentina and Brazil

Example 3.1 (Coffee and wine). Argentina and Brazil each have 10,000 labor hours per month. In Argentina, one pound of coffee takes 2 hours and one bottle of wine takes 4 hours. In Brazil, one pound of coffee takes 1 hour and one bottle of wine takes 5 hours. Which country has an absolute advantage in coffee? Which has a comparative advantage in wine?

Absolute advantage. Brazil needs 1 hour per pound of coffee versus Argentina's 2: Brazil has the absolute advantage in coffee. Argentina needs 4 hours per bottle of wine versus Brazil's 5: Argentina has the absolute advantage in wine. Here each country is better at one good, so absolute and comparative advantage happen to line up; that is a coincidence, not the reason for trade.

Opportunity costs.

Opportunity cost of...	1 lb coffee	1 bottle wine
Argentina	$2/4 = 0.5$ bottle wine	$4/2 = 2$ lb coffee
Brazil	$1/5 = 0.2$ bottle wine	$5/1 = 5$ lb coffee

Comparative advantage. Brazil gives up 0.2 bottle of wine per pound of coffee versus Argentina's 0.5: Brazil has the comparative advantage in coffee. Argentina gives up 2 pounds of coffee per bottle of wine versus Brazil's 5: Argentina has the comparative advantage in wine. Brazil should export coffee and import wine; Argentina should do the reverse. A mutually beneficial price of wine is anywhere between 2 and 5 pounds of coffee per bottle.

The recipe is always the same: (1) hours per unit, or units per hour, for each producer; (2) opportunity cost of each good for each producer, as a ratio; (3) the lower opportunity cost gets the comparative advantage; (4) the trade price lies between the two opportunity costs.

3.6 Applications

3.6.1 Should a star athlete mow his own lawn?

Suppose a star athlete can mow his lawn in 2 hours, and in those 2 hours could instead film a commercial for \$20,000. The teenager next door takes 4 hours to mow the same lawn and could otherwise earn \$40 at a part-time job. The athlete has the absolute advantage in mowing (2 hours versus 4), but his opportunity cost of mowing is \$20,000 against the teenager's \$40. The teenager has the comparative advantage. Any payment between \$40 and \$20,000 makes both better off. The principle applies to individuals exactly as it applies to countries; it is why people specialize in one occupation and buy almost everything else.

3.6.2 Should the United States trade with China?

Replace "Japan" with "China" and "computers" with clothing, furniture, or electronics. If China's opportunity cost of those goods is lower, Americans as a whole gain by importing them and putting U.S. labor into goods where the U.S. has the comparative advantage. Surveyed economists overwhelmingly agree that trade with China makes most Americans better off. They also agree that some Americans who work in industries that compete with those imports are made worse off. The gains exceed the losses, so the winners could in principle compensate the losers, but that does not happen automatically. This is why trade benefits the country as a whole and is still politically contested.

3.7 Summary

Key point

- Trade lets people consume more, and more variety, than they could produce on their own.
- Absolute advantage: fewer inputs. Comparative advantage: lower opportunity cost. Gains from trade rest on comparative advantage. One party can have an absolute advantage in everything, but no one can have a comparative advantage in everything.
- Each party should specialize in its comparative advantage and trade for the other good, at a price between the two opportunity costs.
- The principle applies to individuals and countries alike; it is the basis of the case for free trade.

3.8 Exercises

Exercise 3.1 (Socks in Boston and Chicago). A worker in Boston can make 3 pairs of red socks or 3 pairs of white socks per hour. A worker in Chicago can make 2 pairs of red socks or 1 pair of white socks per hour.

- (a) Which city has the absolute advantage in red socks? In white socks?
- (b) Compute the opportunity cost of a pair of red socks and of a pair of white socks in each city.
- (c) Which city has the comparative advantage in red socks? In white socks?
- (d) If the two cities trade, what is the range of prices for a pair of white socks (measured in pairs of red socks) at which both gain?

Exercise 3.2 (Scones and sweaters). In an hour, an English worker can produce 30 scones or 1 sweater. In an hour, a Scottish worker can produce 40 scones or 2 sweaters. Each country has 40 labor hours per week.

- (a) Draw the PPF of each country (scones on the horizontal axis).
- (b) Which country has the absolute advantage in each good? Which has the comparative advantage in each good?
- (c) If England and Scotland trade, which good does each export? Give the range of prices for a sweater, in scones, at which both gain.
- (d) Suppose instead a Scottish worker could produce 30 scones or 1 sweater per hour. Would there be gains from trade? Explain.

Exercise 3.3 (Wheat and oil). In a day, a worker in Hungary can produce 10 bushels of wheat or 60 barrels of oil. A worker in Slovakia can produce 5 bushels of wheat or 55 barrels of oil.

- (a) Which country has the absolute advantage in wheat? In oil?

- (b) What is the opportunity cost of a bushel of wheat in each country? Of a barrel of oil?
- (c) Which country should export wheat, and which should export oil?
- (d) Slovakia offers 8 barrels of oil per bushel of Hungarian wheat. Does Hungary accept? What about 5 barrels? Would Slovakia offer 12?

Answers to exercises

Exercise 3.1.

- (a) Boston makes more of both per hour (3 vs. 2 red; 3 vs. 1 white): absolute advantage in both.
- (b) Boston: 1 red pair costs 1 white pair, and vice versa. Chicago: 1 red pair costs $1/2$ white pair; 1 white pair costs 2 red pairs.
- (c) Red: Chicago ($1/2 < 1$). White: Boston ($1 < 2$).
- (d) Boston exports white, Chicago exports red. A white pair must sell for more than 1 red pair (Boston's cost) and less than 2 (Chicago's cost).

Exercise 3.2.

- (a) Straight lines. England: from 1,200 scones to 40 sweaters. Scotland: from 1,600 scones to 80 sweaters.
- (b) Absolute advantage: Scotland in both (40 vs. 30 scones; 2 vs. 1 sweaters). Opportunity cost of a sweater: 30 scones in England, $40/2 = 20$ in Scotland; of a scone: $1/30$ vs. $1/20$ sweater. Comparative advantage: Scotland in sweaters, England in scones.
- (c) Scotland exports sweaters, England exports scones. The price of a sweater lies between 20 and 30 scones.
- (d) No. Scotland's opportunity cost of a sweater would be 30 scones, the same as England's. With equal opportunity costs there is no comparative advantage and no price that makes both better off, even though Scotland still has an absolute advantage in scones.

Exercise 3.3.

- (a) Hungary (10 vs. 5 wheat; 60 vs. 55 oil): absolute advantage in both.
- (b) Wheat: $60/10 = 6$ barrels in Hungary, $55/5 = 11$ in Slovakia. Oil: $1/6$ bushel in Hungary, $1/11$ in Slovakia.
- (c) Hungary exports wheat ($6 < 11$); Slovakia exports oil ($1/11 < 1/6$).
- (d) The range is 6 to 11 barrels per bushel. At 8, Hungary accepts (more than the 6 it gets by making oil itself) and Slovakia gains (less than the 11 it costs to grow wheat). At 5, Hungary refuses. Slovakia would not offer 12, since it can grow a bushel itself for 11 barrels.

4 The Market Forces of Supply and Demand

Supply and demand are the two words economists use most, and this chapter is where the tools behind them are built. The model answers four questions: what determines how much buyers want to buy, what determines how much sellers want to sell, how a price emerges from their interaction, and how that price responds when circumstances change. Almost every later chapter uses this machinery, so the time invested here pays off repeatedly.

4.1 Markets and competition

Definition 4.1 (Market). A **market** is a group of buyers and sellers of a particular good or service. The buyers as a group determine the demand for the product; the sellers as a group determine its supply.

Markets take many forms. Some are organized (a stock exchange, a farmers' market); most are not (the market for haircuts in Waltham consists of a few dozen shops and everyone who walks into them). What matters for the model is not the physical setting but how much influence any single participant has over the price.

Definition 4.2 (Competitive market). A **competitive market** has many buyers and many sellers, each so small relative to the market that no one of them can affect the market price. A **perfectly competitive market** adds two assumptions:

- (i) the goods offered by different sellers are identical, and
- (ii) buyers and sellers are so numerous that each one is a **price taker**: they treat the market price as given and decide only how much to buy or sell at that price.

Few real markets meet both assumptions exactly (wheat and other commodities come closest), but the model of perfect competition is the right starting point. It is the simplest case, it captures the essential logic that reappears in more complicated markets, and later chapters relax the assumptions one at a time.

4.2 Demand

4.2.1 Quantity demanded and the law of demand

Definition 4.3 (Quantity demanded). The **quantity demanded** of a good is the amount buyers are willing *and able* to purchase at a given price. Both parts matter: wanting a car is not demand; wanting a car and having the money to buy one at the going price is.

Definition 4.4 (Law of demand). Other things being equal, the quantity demanded of a good falls when its price rises and rises when its price falls.

The phrase *other things being equal* (*ceteris paribus*) is doing real work. The law of demand describes how quantity responds to price when everything else that might influence buyers, such as their incomes, the prices of other goods, and their tastes, is held fixed. Those other things are addressed below under shifts of the demand curve.

4.2.2 The demand schedule and the demand curve

Definition 4.5 (Demand schedule and demand curve). A **demand schedule** is a table listing the quantity demanded at each price, other things equal. A **demand curve** is the graph of that relationship, with price on the vertical axis and quantity on the horizontal axis.

Example 4.1 (A market demand schedule). Table 1 gives the demand schedule for a good sold at prices from \$0 to \$50. The quantity demanded falls by 10 units for every \$10 increase in price, so the schedule can also be written as the equation $Q_d = 50 - P$. Figure 3 plots the six points and connects them.

Price (\$)	Quantity demanded
0	50
10	40
20	30
30	20
40	10
50	0

Table 1: Demand schedule: $Q_d = 50 - P$.

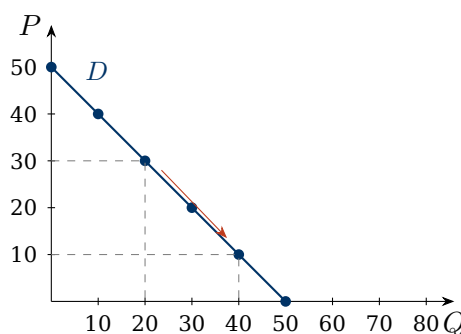


Figure 3: The demand curve. A price cut from \$30 to \$10 raises quantity demanded from 20 to 40: a movement along the curve.

The demand curve slopes downward: that is the law of demand in graphical form. A change in the price of the good moves buyers from one point on the curve to another. It does not move the curve.

4.2.3 Individual demand and market demand

Definition 4.6 (Market demand). The **market demand** for a good is the sum of the individual demands of all buyers in the market. At each price, the market quantity demanded is the sum of the quantities demanded by every buyer. Graphically, the market demand curve is the *horizontal* sum of the individual demand curves.

Example 4.2 (Two buyers). Suppose the market in Table 1 has only two buyers, A and B, with the individual schedules in Table 2. At a price of \$20, A wants 18 units and B wants 12, so the market quantity demanded is 30. Repeating this at every price reproduces the market schedule. In equation form, $Q_A = 30 - 0.6P$ and $Q_B = 20 - 0.4P$, so $Q_d = Q_A + Q_B = 50 - P$.

Price (\$)	Buyer A	Buyer B	Market Q_d
0	30	20	50
10	24	16	40
20	18	12	30
30	12	8	20
40	6	4	10
50	0	0	0

Table 2: Market demand is the horizontal sum of individual demands.

A two-buyer market violates the “many buyers” assumption, but the point holds with two buyers or two million: add quantities across buyers at each price.

4.2.4 Shifts of the demand curve

The demand curve is drawn holding fixed everything other than the good’s own price. When one of those other things changes, the quantity demanded changes *at every price*, and the whole curve moves.

Definition 4.7 (Increase and decrease in demand). An **increase in demand** is a rightward shift of the demand curve: buyers want more at every price. A **decrease in demand** is a leftward shift: buyers want less at every price.

Five variables shift the demand curve. Table 3 lists them with the direction of the shift.

Shifter	Change	Effect on demand
Number of buyers	More buyers enter the market	Increases (shifts right)
Income	Income rises, good is normal	Increases
	Income rises, good is inferior	Decreases (shifts left)
Price of a related good	Price of a substitute rises	Increases
	Price of a complement rises	Decreases
Tastes	Preferences move toward the good	Increases
Expectations	Buyers expect a higher future price or higher future income	Increases (today)

Table 3: Demand shifters. Each row reversed gives the opposite shift.

Definition 4.8 (Normal and inferior goods). A good is **normal** if, other things equal, an increase in income raises the demand for it. A good is **inferior** if an increase in income lowers the demand for it. Most goods are normal; instant noodles, bus rides, and secondhand furniture are common examples of inferior goods.

Definition 4.9 (Substitutes and complements). Two goods are **substitutes** if an increase in the price of one raises the demand for the other (pizza and burgers, Coke

and Pepsi, streaming and the movie theater). Two goods are **complements** if an increase in the price of one lowers the demand for the other (smartphones and apps, printers and ink, tuition and textbooks).

Whether a good is normal or inferior, and whether two goods are substitutes or complements, is an empirical matter about buyers' behavior, not a property of the goods themselves. Frozen pizza may be inferior for one group of buyers and normal for another.

Key point

A change in the good's own price causes a **change in quantity demanded**: a movement along a fixed demand curve. A change in any other determinant causes a **change in demand**: a shift of the entire curve. The rule of thumb: if the variable that changed is on one of the axes, move along the curve; if it is not on either axis, shift the curve.

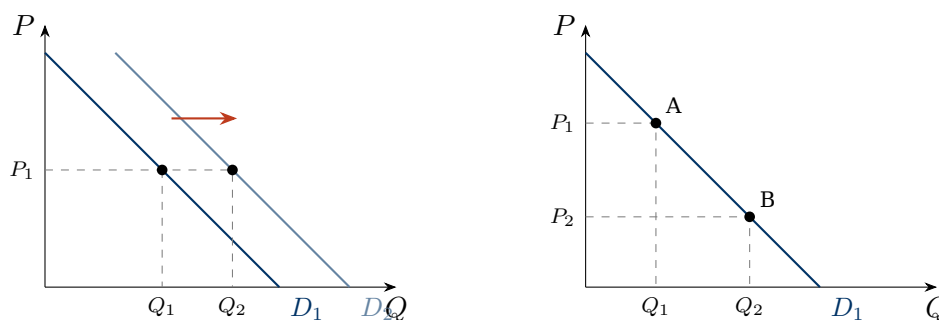


Figure 4: Left: an increase in demand shifts the curve right; at the unchanged price P_1 , buyers want Q_2 instead of Q_1 . Right: a fall in the good's own price moves buyers from A to B along the same curve.

In class

Spotify sells the same subscription at the same price in every country it serves. Draw the demand curve for Spotify subscriptions and show what each event does to it.

- Spotify launches in twenty more countries without changing its price.
- A global recession lowers incomes everywhere.
- Apple Music cuts its price.
- Smartphones become cheaper.
- Spotify starts a campaign promoting local artists in developing countries.
- Prominent economists warn that a recession is coming.
- Spotify eliminates its free tier, so new users must pay the subscription price.

Answers. (a) More buyers: demand shifts right. (b) Income falls; if a subscription is a normal good, demand shifts left. The question of whether Spotify is normal or inferior

has no answer from theory alone; it depends on how subscribers actually respond, and a plausible case can be made either way (people may cut subscriptions, or may substitute cheap streaming for expensive concerts). (c) Apple Music is a substitute; a lower substitute price shifts Spotify demand left. (d) Phones are a complement; cheaper phones shift demand right. (e) Tastes: if the campaign makes the service more attractive, demand shifts right. (f) Expected future income falls; demand shifts left today. (g) For the users affected, the price they face rises from zero to the subscription price. That is a change in the good's own price: a movement along the demand curve, not a shift.

4.3 Supply

4.3.1 Quantity supplied and the law of supply

Definition 4.10 (Quantity supplied). The **quantity supplied** of a good is the amount sellers are willing and able to sell at a given price.

Definition 4.11 (Law of supply). Other things being equal, the quantity supplied of a good rises when its price rises and falls when its price falls.

A higher price makes each unit sold more profitable, so existing sellers produce more and marginal producers find it worth entering. The supply curve therefore slopes upward.

4.3.2 The supply schedule and the supply curve

Definition 4.12 (Supply schedule and supply curve). A **supply schedule** is a table listing the quantity supplied at each price, other things equal. A **supply curve** is the graph of that relationship.

Example 4.3 (A market supply schedule). Table 4 gives the supply schedule for the same good as Table 1. Quantity supplied rises by 15 units for every \$10 increase in price: $Q_s = 1.5P$. Figure 5 plots it.

4.3.3 Individual supply and market supply

Definition 4.13 (Market supply). The **market supply** of a good is the sum of the individual supplies of all sellers. At each price, add the quantities supplied by every seller; graphically, sum the individual supply curves horizontally.

Example 4.4 (Two sellers). If the market in Table 4 has two sellers with $Q_1 = P$ and $Q_2 = 0.5P$, then at \$20 seller 1 offers 20 units and seller 2 offers 10, for a market quantity supplied of 30. Summing at every price gives $Q_s = 1.5P$.

Price (\$)	Quantity supplied
0	0
10	15
20	30
30	45
40	60
50	75

Table 4: Supply schedule: $Q_s = 1.5P$.

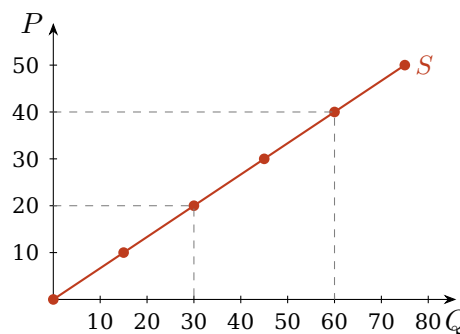


Figure 5: The supply curve. A price rise from \$20 to \$40 raises quantity supplied from 30 to 60: a movement along the curve.

4.3.4 Shifts of the supply curve

Definition 4.14 (Increase and decrease in supply). An **increase in supply** is a rightward shift of the supply curve: sellers offer more at every price. A **decrease in supply** is a leftward shift: sellers offer less at every price.

Shifter	Change	Effect on supply
Input prices	Wages, raw materials, energy become cheaper	Increases (shifts right)
Technology	A cost-saving improvement in production	Increases
Number of sellers	More firms enter the market	Increases
Expectations	Sellers expect a higher future price (storable goods)	Decreases today (shifts left)

Table 5: Supply shifters. Each row reversed gives the opposite shift.

Input prices and technology work through cost: anything that lowers the cost of producing a unit makes production more profitable at every price, so sellers offer more. The expectations row runs the opposite way from the demand table: a seller who expects a higher price next month holds inventory back today, reducing current supply.

Common mistake

Two errors recur on exams.

- Describing a supply shift as “up” or “down.” A supply curve that moves up is a *decrease* in supply (less offered at each price). Always describe shifts as right (increase) or left (decrease), reading horizontally at a fixed price.
- Putting a demand shifter on the supply curve. A fall in the price of orange juice affects the *demand* for apple juice (they are substitutes for buyers). It does nothing to the supply of apple juice, because it does not change the cost

of making apple juice.

Key point

A change in the good's own price causes a **change in quantity supplied**: a movement along the supply curve. A change in input prices, technology, the number of sellers, or expectations causes a **change in supply**: a shift of the curve.

4.4 Supply and demand together

4.4.1 Equilibrium

Definition 4.15 (Equilibrium). A market is in **equilibrium** when the price has reached the level at which quantity supplied equals quantity demanded. That price is the **equilibrium price** (or market-clearing price) and the common quantity is the **equilibrium quantity**. On the graph, equilibrium is the intersection of the supply and demand curves.

Example 4.5 (Equilibrium from the schedules). Table 6 puts the demand and supply schedules side by side. At \$20, quantity demanded and quantity supplied are both 30. That is the only price at which they match, so the equilibrium is $P^* = 20$, $Q^* = 30$. Figure 6 shows the same thing graphically.

Price (\$)	Q_d	Q_s	$Q_s - Q_d$
0	50	0	-50
10	40	15	-25
20	30	30	0
30	20	45	25
40	10	60	50
50	0	75	75

Table 6: Negative entries in the last column are shortages; positive entries are surpluses.

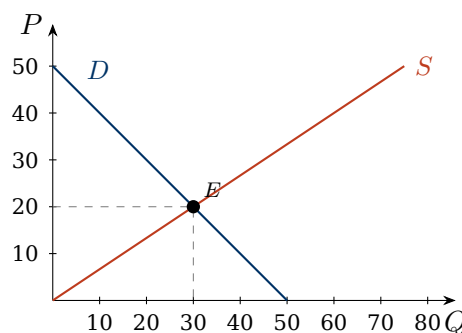


Figure 6: Equilibrium at $P^* = 20$, $Q^* = 30$.

4.4.2 Surplus and shortage

What if the price is not at its equilibrium level? The last column of Table 6 answers this.

Definition 4.16 (Surplus). A **surplus** (excess supply) exists when the quantity supplied exceeds the quantity demanded. It arises when the price is above the equilibrium price.

Definition 4.17 (Shortage). A **shortage** (excess demand) exists when the quantity demanded exceeds the quantity supplied. It arises when the price is below the equilibrium price.

Example 4.6 (Surplus at \$40, shortage at \$10). At $P = 40$, sellers offer 60 units but buyers want only 10: a surplus of 50. Sellers holding unsold inventory cut the price. As the price falls, quantity demanded rises (down along D) and quantity supplied falls (down along S), so the surplus shrinks until it disappears at \$20.

At $P = 10$, buyers want 40 units but sellers offer only 15: a shortage of 25. Too many buyers chase too few units, and sellers can raise the price without losing customers. As the price rises, quantity demanded falls and quantity supplied rises, until the shortage is gone at \$20. Figure 7 shows both cases.

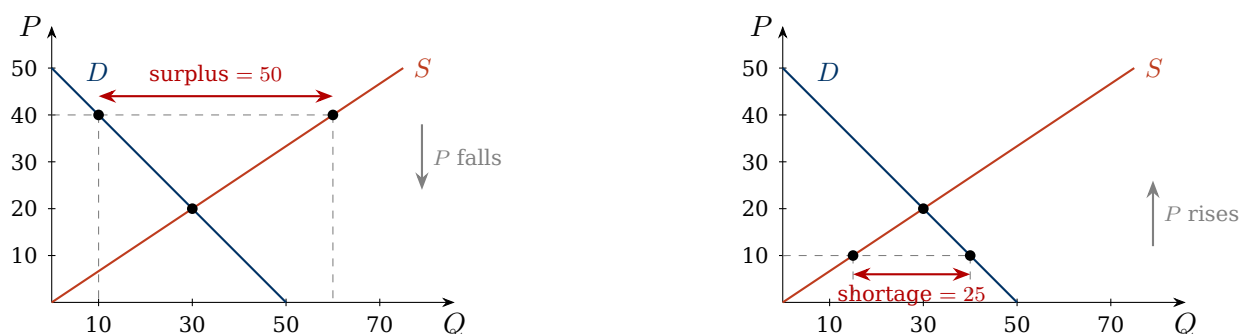


Figure 7: Left: at \$40 a surplus of 50 pushes the price down. Right: at \$10 a shortage of 25 pushes it up. Both end at \$20.

Key point

Law of supply and demand. The price of any good adjusts to bring the quantity supplied and the quantity demanded into balance. Once the market reaches equilibrium, there is no further pressure on the price in either direction. How fast this happens varies across markets, but in most well-functioning markets surpluses and shortages are temporary.

4.4.3 Equilibrium from equations

When demand and supply are given as equations, equilibrium is the price that solves $Q_d = Q_s$.

Example 4.7 (Solving for equilibrium). Let $Q_d = 1200 - 50P$ and $Q_s = 200 + 50P$. Setting them equal,

$$1200 - 50P = 200 + 50P \implies 1000 = 100P \implies P^* = 10.$$

Substituting back, $Q^* = 1200 - 50(10) = 700$; checking with supply, $200 + 50(10) = 700$. Always verify with the second equation, since an arithmetic slip in the first will show up as a mismatch.

The schedules of Table 6 work the same way: $50 - P = 1.5P$ gives $P^* = 20$ and $Q^* = 30$.

To find a surplus or shortage at a given price, evaluate both equations at that price and subtract. At $P = 14$ in the example, $Q_d = 500$ and $Q_s = 900$: a surplus of 400. At $P = 6$, $Q_d = 900$ and $Q_s = 500$: a shortage of 400.

4.5 Three steps to analyzing a change in equilibrium

An event that shifts demand or supply changes the equilibrium. To predict how, follow three steps.

Key point

1. Decide whether the event shifts the demand curve, the supply curve, or both.
2. Decide whether each affected curve shifts right or left.
3. Draw the diagram and compare the new equilibrium with the old one: what happened to P and Q ?

Step 1 requires knowing the shifter lists (Tables 3 and 5). Ask: does the event change what buyers want at a given price, or what it costs sellers to produce? Step 2 requires the direction. Step 3 is mechanical once the first two are right.

Example 4.8 (The electric car market: three cases). Assume the market for electric cars starts in equilibrium at (Q_1, P_1) .

Case A: the price of oil rises. Gasoline cars run on oil; electric cars do not. For a buyer, a gasoline car and an electric car are substitutes, and the higher oil price makes gasoline cars more expensive to own. Step 1: demand shifts (this is a related-good price on the buyer side; the cost of building an electric car is unchanged). Step 2: demand shifts right. Step 3: at the old price there is a shortage, the price is bid up, and the new equilibrium has a higher price and a higher quantity (Figure 8, left).

Case B: engine technology improves. A cheaper way to build the motor lowers production cost. Step 1: supply shifts. Step 2: supply shifts right. Step 3: at the old price there is a surplus, the price falls, and the new equilibrium has a lower price and a higher quantity (Figure 8, center).

Case C: both at once. Both curves shift right. Both shifts raise quantity, so Q rises unambiguously. The demand shift pushes the price up and the supply shift pushes it down, so the net effect on P depends on which shift is larger. Without that information, the effect on price is **ambiguous** (Figure 8, right, drawn with the supply shift larger, so that P falls).

The same logic covers every combination. Table 7 summarizes. When only one curve shifts, both P and Q are determined. When both shift, one of the two is determined and the other is ambiguous.

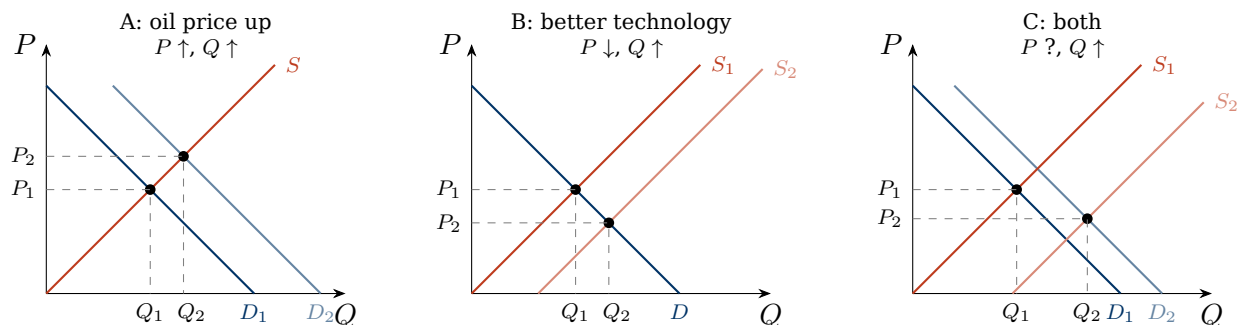


Figure 8: The electric car market. In case C the direction of P depends on the relative sizes of the shifts.

	No change in supply	Supply increases	Supply decreases
No change in demand	P same, Q same	$P \downarrow, Q \uparrow$	$P \uparrow, Q \downarrow$
Demand increases	$P \uparrow, Q \uparrow$	P ambiguous, $Q \uparrow$	$P \uparrow, Q$ ambiguous
Demand decreases	$P \downarrow, Q \downarrow$	$P \downarrow, Q$ ambiguous	P ambiguous, $Q \downarrow$

Table 7: Effects of shifts on equilibrium price and quantity.

Note

When the problem says both curves shift but not by how much, the correct answer for the ambiguous variable is “cannot be determined,” not a guess. If the problem does specify which shift is larger, the ambiguity disappears: in case C, a larger demand shift means P rises, a larger supply shift means P falls.

In class

For each statement, draw the shift in the named market that is consistent with it and identify the shifter.

- Healthcare: “When there is an influx of immigrants, the price of healthcare rises.”
- Electric cars: “When the price of electric cars is expected to rise in the near future, the current price of electric cars rises.”
- Laptops, then used desktops: “When a technological advance improves the features of laptops, the price of laptops rises and the price of used desktops falls.”

Answers. (a) More buyers: demand for healthcare shifts right; P and Q rise. (b) Two forces point the same way. Buyers who expect a higher price buy now (demand right). Sellers who expect a higher price hold cars back (supply left). Either alone raises today’s price; together they raise it for sure and leave Q ambiguous. Drawing only the demand shift is enough to match the statement. (c) This is a two-market problem. In the laptop market, the advance improves the product rather than lowering its cost, so it works through tastes: demand for laptops shifts right and the laptop price rises. (If

instead the advance lowered production cost, supply would shift right and the price would fall, which contradicts the statement; the statement tells you which interpretation is meant.) In the used-desktop market, laptops are a substitute, and the better laptop draws buyers away: demand for used desktops shifts left, and their price falls.

4.6 How prices allocate resources

Chapter 1 claimed that markets are usually a good way to organize economic activity. This chapter shows the mechanism. In a market economy nobody decides how many electric cars to build or who gets them. The price does. When demand rises, the higher price signals producers to build more and rations the cars among the buyers who value them most; when a new technology cuts costs, the lower price spreads the gain to buyers and pulls resources into the industry. Prices are the signals that guide every decision in the economy, and the equilibrium price is the one that makes those decisions consistent: what buyers choose to consume equals what sellers choose to produce.

That is the idea behind the invisible hand. The rest of the course studies when this mechanism works well and when it does not.

4.7 Exercises

Exercise 4.1. The market for pizza slices on campus has the following schedules.

Price (\$)	Q_d	Q_s
1	120	0
2	100	40
3	80	80
4	60	120
5	40	160
6	20	200

- Find the equilibrium price and quantity.
- At a price of \$5, is there a surplus or a shortage, and how large? What happens to the price?
- At a price of \$2, is there a surplus or a shortage, and how large?
- A new dorm opens and quantity demanded rises by 60 slices at every price. Write the new demand schedule and find the new equilibrium.

Exercise 4.2. Demand and supply for a good are $Q_d = 1200 - 50P$ and $Q_s = 200 + 50P$.

- Find the equilibrium price and quantity.
- Find the surplus or shortage at $P = 16$ and at $P = 4$.
- Income rises and the good is normal; demand becomes $Q_d = 1500 - 50P$. Find the new equilibrium. Then explain, using the words *shortage* and *movement along the supply curve*, how the market gets there.

Exercise 4.3. Apply the three-step method to each event and state the effect on equilibrium price and quantity.

- (a) A hurricane destroys a large share of the Hudson Valley apple crop. Analyze the market for apples. Then analyze the market for apple cider.
- (b) A scientific report finds that fish contain dangerous levels of mercury. At the same time, the price of diesel fuel used by fishing boats falls. Analyze the market for fish.

Exercise 4.4. (a) Honey and jam are substitutes. The price of honey rises. What happens in the market for honey? What happens in the market for jam, and to the price of jam?

- (b) Retirees are the main buyers of housing in Florida. Which of the following shifts the demand curve for Florida housing to the right: (i) the aging of the baby boom generation, which raises the number of people reaching retirement; (ii) higher homeowners' insurance premiums; (iii) high current real-estate prices; (iv) stricter building codes that raise construction costs. Explain what each of the others does instead.

Answers to exercises

Exercise 4.1. (a) $Q_d = Q_s = 80$ at $P = 3$, so $P^* = 3$, $Q^* = 80$. (b) At \$5, $Q_s = 160$ and $Q_d = 40$: a surplus of 120. Sellers with unsold slices cut the price; it falls toward \$3. (c) At \$2, $Q_d = 100$ and $Q_s = 40$: a shortage of 60. The price rises. (d) New Q_d at prices \$1 to \$6: 180, 160, 140, 120, 100, 80. It equals Q_s at \$4 ($120 = 120$), so the new equilibrium is $P^* = 4$, $Q^* = 120$. More buyers, demand right, P and Q both up.

Exercise 4.2. (a) $1200 - 50P = 200 + 50P$ gives $P^* = 10$, $Q^* = 700$. (b) At $P = 16$: $Q_d = 400$, $Q_s = 1000$, surplus of 600. At $P = 4$: $Q_d = 1000$, $Q_s = 400$, shortage of 600. (c) $1500 - 50P = 200 + 50P$ gives $P^* = 13$, $Q^* = 850$ (check: $200 + 650 = 850$). At the old price of \$10, quantity demanded is now $1500 - 500 = 1000$ while quantity supplied is still 700: a shortage of 300. The shortage pushes the price up. As it rises, sellers move up along the unchanged supply curve from 700 toward 850, and buyers move up along the new demand curve from 1000 toward 850. The shortage closes at \$13.

Exercise 4.3. (a) Apples: the hurricane reduces the number of sellers able to supply, and raises the cost of supplying, apples. Supply shifts left; P rises, Q falls. Cider: apples are an input, and their price has risen. Supply of cider shifts left; the price of cider rises and the quantity falls. (b) Fish: the report changes tastes; demand shifts left. Cheaper diesel lowers an input price; supply shifts right. Both shifts lower the price, so P falls. The demand shift lowers Q and the supply shift raises it, so Q is ambiguous.

Exercise 4.4. (a) Honey: a change in the good's own price is a movement along the honey demand curve, not a shift; quantity of honey demanded falls. Jam: honey is a substitute and has become more expensive, so demand for jam shifts right; the price and quantity of jam both rise. (b) Only (i): more buyers shift demand right. (ii) Insurance is a complement to owning a home; higher premiums shift demand left. (iii) The current price of housing is on the vertical axis; it determines a point on the

demand curve, not its position. (iv) Construction cost is a seller's cost; it shifts supply left, not demand.